AUtoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 2: Model Testing, Validation and Refinement

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Status: development results; the one-time final test has not been run

The final test needs a fresh, locked benchmark that does not exist yet (its protocol is declared in Section 2). Every result below comes from held-out validation questions the models were never trained on, and each table says so. I make no final verdict and no deployment claim.

1. Summary of Results

The baseline is Nemotron 3.5 Lightning's untouched base model; the model is the same base with my fine-tuned adapter (1,600 training records, checkpoint 400). My PM Plan threshold is at least 10 percentage points more correct, fully supported answers than the base, with fewer than 2% of answers containing an unsupported claim. On the 24-question development probe both gave 4 fully supported answers (17 per 100 questions): a difference of 0 points, so the threshold is not met. On the 15 questions I reviewed, the adapter gave 4 strict answers and the base 2 (27 vs. 13 per 100), a 13-point difference that is not statistically reliable ($p = 0.5$). The adapter put an unsupported claim in 19 of 24 answers (79 per 100) and never declined. In business terms, an analyst would get about 17 usable answers per 100 questions and would have to catch about 79 answers with an invented or wrong policy detail. Verdict on development evidence: the model does not meet the threshold and would not be deployed. The final test will confirm or overturn this.

2. Experimental Setup

Frozen configuration. NVIDIA Nemotron 3.5 Lightning 30B-A3B, BF16, revision a9904d2. Adapter: LoRA rank 16 trained on 1,600 records, checkpoint 400 (adapter SHA-256 4aaca5c5…). Decoding: greedy (no sampling), at most 512 new tokens, reasoning mode off, seed 20261006. The base condition is the same weights with the adapter switched off. Development runs gave the model only the question. On October 14 I changed the final-test input: both conditions receive the same official source passage with the question, so the test measures whether fine-tuning improves reading and citing the right rule rather than recalling a manual from memory. No retrieval system is involved.

Split and test protection. Records are split by document (1,684 train / 124 validation / 191 test). All tuning used validation records only: checkpoints were chosen by validation loss, and the 24-question probe and the Week 8 review used validation questions. The 191 historical test records were used for diagnostics in September, so I no longer treat them as untouched. The final test will use fresh questions from official documents outside all three splits. An October 14 audit found no exposure evidence for 11 of 12 candidate documents and flagged one pharmacy chapter, which is not used unless its review clears it. The questions are written by a local AI model and checked against the source, not written by me. Membership, prompts, settings and scoring rules are locked before the single run, and nothing is tuned afterward.

Metrics. Primary: factual accuracy on answerable questions; an answer counts only if it is correct, supported by the source and free of unsupported claims (strict accuracy). Secondary: unsupported-claim rate, correct abstention on unanswerable questions and citation correctness. They reflect the business asymmetry: a confident wrong policy statement can lead to a wrong billing decision, while a refusal only costs analyst time.

Changes from the PM Plan, declared October 18 before any final output exists. (1) The PM Plan committed to human-judged accuracy. I am changing this because I will not do a blind human-rating sample; instead, two LLM judges will score the full output set, which broadens coverage while keeping the scoring consistent. This is a trade-off, not equivalent human validation. The judges are a local Qwen model (the October development judge) and Claude Sonnet 5.5 through the Claude Code CLI, authenticated through the Hermes Direct Claude Code SDK subscription; a fixed script runs fresh sessions without tools or access to the answer key. Neither is the model under test. An answer passes only if both judges rate it correct, supported and free of unsupported claims; I report each judge separately, with raw agreement and Cohen's kappa. Limit: the Qwen family also drafted the training records, so the two judges are not fully independent of the project, which is why the Claude judge must agree too. (2) The October check against my review (Section 5) showed the AI judge matched me on correctness, support and unsupported claims but not on citations (3 of 12), so citation correctness is checked in code against the supplied page and the expected locator, not by any LLM.

Final-test protocol, locked and fingerprinted before the single run

Element	What is fixed in advance
Benchmark	Fresh questions written from official documents outside all three splits by a local model and checked against the source page; about 120 answerable and 30 unanswerable questions planned (the accepted set sets the final counts). The flagged pharmacy source is excluded unless its review clears it.
Conditions	Nemotron adapter (checkpoint 400) and its untouched base, each given the same source passage and question; greedy decoding, 512 new tokens, seed 20261006. One run; failed outputs stay in the denominator; no retries or tuning afterward.
Scoring	Strict accuracy by two LLM judges (consensus rule above); citations checked in code; on unanswerable questions a correct answer is a decline.
Robustness	Each passage cut to its first 60% of characters; questions whose evidence survives the cut and questions whose evidence is removed are scored separately.
Subgroups	Document family, and answerable versus unanswerable questions.
Thresholds	From the PM Plan: +10 points over base on answerable accuracy; at least 95% correct citations; under 2% unsupported claims; at least 90% correct abstention. Safety routing and general-task regression are reported as not tested unless the locked protocol adds them.

3. Iteration Narrative

The run log (16 training runs and four development experiments) gives this arc. Failed attempts stay in the record because they ruled things out.

When	What I tried	Expected	What happened	What I concluded
Sep 16–18	First Muse fine-tunes	Adapter beats base	Used earlier data-generation and training methods that I later amended	Excluded the runs; all later runs use the amended data, split and procedure.
Sep 29–Oct 1	16 runs: 4 families × 200 / 509 / 750 / 1,600 records, same validation set	More data lowers validation loss	True for three families; Muse flat after 509. From 509 records up, every family overfit after 1–2 epochs. Early Nemotron and Ornith attempts failed (one on an unsupported warm-up setting)	Use 1,600 records and the early checkpoint. Fixed the training runner and reran all eight runs from scratch.
Oct 6–7	Week 8 error review: 15 questions × base and adapter (Nemotron)	Adapter more accurate	Strict accuracy 2/15 → 4/15; 0 of 30 citations correct; invented rules most common	Citations are format only; question-only input invites recall errors.
Oct 7	Refinement: added an instruction to use only verifiable source facts (8 questions, 2 documents)	Fewer unsupported claims	Base swapped answers for “cannot verify” (correct 3/8 → 0/8); adapter unchanged at 4/8 and still cited wrong sources on 8/8	Prompt wording alone does not fix grounding. Not promoted.
Oct 8	Cross-family probe: 192 outputs scored by a local AI judge	A fair comparison	The first judge counted refusals as supported answers, so I withdrew it; the corrected judge requires a real answer	Three adapters tie at 4/24; selected Nemotron (Milestone 1).
Oct 14	Changed the final-test input to supply the source passage	Isolates what fine-tuning adds	Evaluation harness built; passed separate specification and code-quality reviews	Ready once the fresh benchmark is locked.

What the failures ruled out: the early data methods, longer training (more epochs overfit), prompt-only fixes and refusal-friendly scoring. What remains unproven is whether fine-tuning helps when the source is supplied, which is exactly what the final test measures.

4. Test Results (Development; Final Test Pending)

The required single test has not been run, so Table 1 gives the development equivalent: the same 24 validation questions (9 documents), the same settings and metric, base versus model versus threshold. The final table will have the same rows, computed once on the locked benchmark, with answerable and unanswerable questions reported separately so a pooled score cannot hide an abstention failure.

Table 1. Baseline, model and threshold on the same development questions

Measure	Untouched base	Adapter (model)	Threshold	Development verdict
Fully supported answers, 24 questions (AI-judged)	4/24 (17%)	4/24 (17%)	+10 points over base	Not met (+0)
Strict accuracy, 15 questions (my review)	2/15 (13%)	4/15 (27%)	+10 points over base	+13 points; not reliable (p = 0.5)
Answers with an unsupported claim, 24 questions	17/24 (71%)	19/24 (79%)	Under 2%	Not met
Citation points to the correct page, 15 questions	0/15	0/15	At least 95%	Not met

Measure	Untouched base	Adapter (model)	Threshold	Development verdict
Correct abstention on unanswerable questions	Not measurable	Not measurable	At least 90%	Not tested: no unanswerable development questions

Table 2. Confusion-matrix equivalent: outcomes on the 24 answerable development questions

Outcome	Untouched base	Adapter
Answered, fully supported	4	4
Answered, but wrong, partly wrong or unsupported	17	20
Declined to answer (unnecessary refusal)	3	0
Generation failed	0	0
Unanswerable questions: correctly declined / answered unsafely	None in development set	None in development set

Business units. Per 100 analyst questions like these, either version would give about 17 answers an analyst could use as drafted. The adapter would give about 79 answers with an invented or wrong detail (the base about 71) and would never say it cannot answer. Manual lookup takes about 84 seconds per simple question, but reviewing and correcting 79 flawed drafts would cost more than 17 usable drafts save, so the development result supports no time-saving claim.

5. Error and Robustness Analysis

The failure modes below come from my Week 8 review and were re-checked on the 24-question probe, both on the frozen candidate.

Table 3. Named failure modes with counts (one label per output)

Failure mode	Week 8 review: base / adapter (15 each)	Probe, AI judge: base / adapter (24 each)	What it suggests
Invented rule: states a rule the source does not contain	9 / 10	11 / 4	Input and data change: supply the source passage; add unanswerable training examples.
Wrong value: wrong number, date or code	3 / 0	6 / 15	Data change: check effective dates and editions. Scope rule: show the quoted source value beside the answer.
Wrong citation on an otherwise correct answer	3 / 5	0 / 2	Scope rule: insert the citation from the supplied passage instead of letting the model write it.
Abstention problem: declined when the answer was available	0 / 0	3 / 0	Data change: teach when to decline, using unanswerable examples.
Missing evidence: answer without the needed support	0 / 0	0 / 1	Input change: supply the source passage.
No error	0 / 0	4 / 2	—

The review also marked “failed to decline” on 10 base and 6 adapter answers. The two sources disagree on the label for half the shared outputs (6 of 12), mostly “wrong value” versus “invented rule”, but they agree that nearly every adapter error is a confident wrong statement rather than a refusal. Across all 150 judged probe outputs (all families), the AI judge found 57 wrong values, 32 abstention problems, 26 invented rules, 8 wrong citations and 26 answers with no error.

Check on the measurement itself. On 12 Nemotron outputs scored by both the AI judge and my review, they agreed on correctness, evidence support, unsupported claims and the overall pass in 12 of 12 cases, but on citation validity in only 3 of 12. My review was AI-assisted (I saw the judge’s suggestion and confirmed all 180 ratings), so I do not reuse those ratings for the final test. The final test uses two LLM judges and a code check for citations (Section 2).

Robustness observation: reworded instructions. On October 7 I reran 8 development questions with a rewritten instruction (use only facts you can verify). For the frozen adapter, correct answers stayed at 4/8 and wrong citations at 8/8, so the measure moved 0 points. The base model was fragile: correct answers fell from 3/8 to 0/8 (−38 points) as it switched to refusing. This was run as a refinement attempt, judged by the AI judge, on only 8 questions, so it is weak evidence; I report it because it shows how far the primary measure moves under a realistic prompt change.

Predeclared final robustness check (not yet run). Each final question is also run with the source passage cut to its first 60% of characters, a realistic loss of context. Questions whose evidence survives the cut and questions whose evidence is removed are scored separately. The expected behavior is unchanged accuracy when the evidence survives and a correct refusal when it is removed. This evaluates the frozen model and does not reopen tuning.

6. Subgroup Performance

The segments my decision affects are the two document families an analyst consults (Administrative Code rules and Provider Manual billing chapters) and answerable versus unanswerable questions.

Table 4. Development results by segment (24-question probe, AI-judged)

Segment	Base: fully supported	Adapter: fully supported	Adapter: unsupported claims
Administrative Code (12 questions)	2/12	1/12	11/12
Provider Manual (12 questions)	2/12	3/12	8/12
Unanswerable questions	Not in development set	Not in development set	—

Gaps as found: the adapter did slightly better on Provider Manual questions and worse on Administrative Code questions, where 11 of 12 answers carried an unsupported claim. With 12 questions per group a two-answer gap is within chance, and Gemma's adapter shows the opposite pattern (3/12 Administrative Code vs. 1/12 Provider Manual), so I read this as noise rather than a stable gap. What I would do: the final benchmark reports each document family with its own denominator, and if Administrative Code accuracy trails, I would limit model drafting to Provider Manual questions. My Week 8 review covered only Provider Manual questions, which is itself a coverage gap.

NIST AI RMF, tied to these results. Measure: a 79% unsupported-claim rate fails the framework's "valid and reliable" characteristic, and 0 of 30 correct citations fails "accountable and transparent", because a reviewer cannot trace the answer to its rule. Manage: these results trigger the deployment block in my plan, so the model stays in testing. Govern: an analyst must approve every answer, and a named owner tracks manual editions. Map: the harm is wrong billing guidance reaching providers, which can delay or deny services to Medicaid members.

EU AI Act. The system is used in Alabama, so the Act does not apply directly. If it did, a tool that evaluates people's eligibility for public benefits would be high-risk (Annex III). My scope excludes eligibility decisions and keeps a human reviewer, which should keep it outside that category, though transparency duties for AI-generated text could still apply.

7. Limitations and Deployment Readiness

- **Population and time validity.** Results cover Alabama Medicaid's public Administrative Code and Provider Manual as downloaded September 14, 2026. The manuals are reissued in dated editions (the corpus holds April and July 2026 files), so an adapter trained on one edition goes stale when the next appears. The development questions came from 9 documents and were all answerable; nothing here speaks to other states or to unanswerable questions.
- **Remaining failure modes.** Invented rules and wrong values, wrong citations and no refusals (Section 5).
- **Why the final test is not done.** The fresh benchmark must be generated, source-checked and locked first (protocol in Section 2), and the one flagged source must be reviewed. With about 30 unanswerable questions, abstention will be reported with its interval rather than as a proven 90% rate. LLM judges can share blind spots, so I report their agreement, check citations in code, and will not call the scores human-judged.
- **Operational requirements.** One GPU with about 64 GB available for the model (63.2 GB allocated after loading on a 96 GB RTX PRO 6000 Blackwell); median answer time 2.9 seconds (90th percentile 3.4 seconds); the source passage retrieved and passed in with each question; analyst approval before any answer is used; and an audit log of question, passage, answer and reviewer decision.

Table 5. Proposed monitoring plan for any pilot

What I would monitor	How	Trigger and response
Unsupported-claim rate	An analyst checks a random 25 answers each week	Above 2% two weeks running: stop drafting and re-evaluate
Citation correctness	Same weekly sample, checked against the cited page	Below 95%: switch to inserted citations and re-evaluate
Abstention	Five seeded unanswerable questions each week	Fewer than 90% declined: block use
Source editions	Compare manual edition dates with the training snapshot monthly	New edition of a covered chapter: re-run the evaluation; retrain only if accuracy drops
Latency	Log the answer time of every request	90th percentile above 10 seconds: capacity review

Readiness: not ready. A pilot requires passing every threshold on the final test. Autonomous policy decisions stay out of scope even then.

8. Repository

Course repository (public): github.com/porespellar/INAI6986_project_deliverables

Development notebook: notebook/current_evidence_replay_20261018.ipynb

The notebook runs top to bottom offline with zero errors and recomputes every development number in this report from the stored records, after checking each input's fingerprint. It does not rerun model generation. A final-test notebook that reproduces the reported test result from the locked outputs and ratings will follow the one-time test.

AI Use Disclosure

I used Hermes Agent with GPT-6 Luna and GPT-6.1 Sol (October 1–16) and Claude Opus 5.5 and Sonnet 5.5 (October 18) to draft and revise this report and to write the evaluation, analysis and document-building code. Local Qwen models generated training records and the advisory development scores. I designed the evaluation, chose the candidate and thresholds, made the Week 8 review ratings, and decided to change the final-test input and to score it with two LLM judges instead of a human sample. I checked the reported figures against the notebook output and remain responsible for the content.

Appendix A — Development Results for All Four Families

Same 24 validation questions, question-only input, greedy decoding, AI-judged. Failed generations count as not correct. Paired exact McNemar tests compare each adapter with its own base.

Model	Condition	Fully supported (95% CI)	Unsupported claims	Abstained	Failed	Paired p
Nemotron	Untouched base	4/24 (7%–36%)	17/24	3	0	
Nemotron	Adapter	4/24 (7%–36%)	19/24	0	0	1.0
Gemma	Untouched base	0/24 (0%–14%)	10/24	14	0	
Gemma	Adapter	4/24 (7%–36%)	18/23	0	1	0.125
Ornith	Untouched base	1/24 (1%–20%)	0/24	23	0	
Ornith	Adapter	4/24 (7%–36%)	18/23	0	1	0.375
Muse	Untouched base	0/24 (0%–14%)	0/8	8	16	
Muse	Adapter	0/24 (0%–14%)	—	—	24	1.0

Appendix B — Evidence Files

Claim	File in the source-of-record repository
Predeclared probe: questions, settings, selection rule	artifacts/milestone2_completion/development/shortlist_v1/frozen/contract.json
192 counted outcomes and judgments	artifacts/milestone2_completion/development/shortlist_v1/verification/development_v2_counted_outcomes_20261008.jsonl
Week 8 review (my ratings, revision 11)	artifacts/milestone2_completion/review/batch_b/verification/submitted_revision_000011/
Refinement experiment	artifacts/milestone2_completion/development/batch_b/refinement_v1/outputs/
Final-test harness reviews	Runtime code fingerprint b4c06b7f5a85770a…; specification and code-quality reviews dated October 14
Exposure audit of fresh sources	October 14 audit receipt (12 documents; 11 with no exposure evidence; 1 flagged)
All derived numbers in this report	artifacts/milestone2_completion/class_package_20261018/development_analysis_20261018.json